Moving Horizon Estimation With Variable Structure Interacting Multiple Model for Surrounding Vehicle States in Complex Environments

Haikuan Lu, Ping Wang, Ting Qu, Hong Chen, *Fellow, IEEE*, Lin Zhang, and Yunfeng Hu

***Abstract*— Motion prediction of surrounding vehicles in complex environments is essential for autonomous vehicle trajectory planning. Accurate motion prediction requires accurately estimating the state information of the surrounding vehicles. For this purpose, a moving horizon estimation with interacting multiple model (IMM-MHE) algorithm is first proposed here. The algorithm can match multiple vehicle maneuvers, but also fully utilizes the historical information obtained during the driving process, achieving a high estimation accuracy. Second, a moving horizon estimation with variable structure interacting multiple model (VSIMM-MHE) framework is designed. Time-domain adaptation is proposed to solve the problem that the fixed time domain of some models cannot be filled due to model activation and elimination. A new interaction method is proposed to solve the problem that models cannot interact because the starting timesteps of their time domains are different. The proposed framework reduces not only the computational burden, but also the final estimation error caused by the model not matching the current maneuver. Third, based on a model set consisting of different kinds of intention models, a VSIMM-MHE algorithm is proposed. This algorithm introduces residual information into the model classification method, reducing the dependence on the accuracy of the model probabilities. It can not only accurately estimate the state information of surrounding vehicles in a complex environment, but also identify the model that best matches the current maneuver and effectively predict the motion trajectories of surrounding vehicles through model probabilities. Finally, joint simulation with SCANeR studio, Carsim and Simulink and hardware-in-the-loop experiment demonstrate the effectiveness of not only the two proposed estimation algorithms but also the motion prediction of surrounding vehicles using the model probabilities in the VSIMM-MHE algorithm.**

***Index Terms*— Surrounding vehicle state estimation, moving horizon estimation, variable structure multiple model estimation, interacting multiple model Kalman filter, motion prediction.**

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NOMENCLATURE

$\hat{x}_0^{(i)}$	Priori estimate of each model's starting state at $k < N_e$.
$\hat{x}_k$	The fused state estimate at k timestep.
$\hat{x}_k^{(i)}$	State estimate of each model at k timestep.
$\hat{x}_{k-Ne}^{(i)}$	Priori estimate of each model's starting state $k \geq N_e$.
$\lambda_k^{(i)}$	The variable used in model classification.
$\mathbb{M}_a$	The model set of all models adjacent to the principle models.
$\mathbb{M}_d$	The set of discardable models.
$\mathbb{M}_k$	The model set at k timestep.
$\mathbb{M}_m$	The set of models with smaller probabilities in $\mathbb{M}_d$.
$\mathbb{M}_n$	Intersection of $\mathbb{M}_a$ and $\overline{\mathbb{M}}_k$.
$\mathbb{M}_u$	The model set consisting of unlikely models.
$\mu_k^{(i)}$	Model probability of each model.
$\omega_k^{(i)}$	Process noise of the system.
$\overline{\mathbb{M}}_k$	The complement of $\mathbb{M}_k$.
$\overline{P}_0^{(i)}$	Covariance matrix of each model's starting state after an interaction at $k < N_e$.
$\overline{P}_{k-Ne}^{(i)}$	Covariance matrix of each model's starting state after an interaction at $k \geq N_e$.
$\overline{x}_0^{(i)}$	Priori estimate of each model's starting state after an interaction at $k < N_e$.
$\overline{x}_{k-Ne}^{(i)}$	Priori estimate of each model's starting state after an interaction at $k \geq N_e$.
π	Transition probability matrix.
π_{ji}	Transition probability from $m^{(j)}$ to $m^{(i)}$.
$\tilde{y}_k^{(i)}$	Residual of each model.
$\upsilon_k^{(i)}$	Measurement noise of the system.
$a_{x,k}$	Longitudinal acceleration at k timestep.
$a_{y,k}$	Lateral acceleration at k timestep.
$d_{pre,k}$	The distance between the vehicle and the pedestrian or obstacle.

$E_k^{(i)}$	External input of the system.
F_i	State transfer matrix of the system.
H_i	Observation matrix of the system.
k	Timestep.
$K_{LQR}^{(VT)}$	LQR gain for the VT model.
$L_k^{(i)}$	Likelihood function of each model.
L_{oc}	The set of models within the current cycle with time domain length N_e.
$m^{(i)}$	The ith model.
N_e	Time-domain length.
N_e^{max}	Maximum time-domain length.
N_{um}	The number of models within the current cycle with time domain length N_e.
$P_0^{(i)}$	Covariance matrix of each model's starting state at $k < N_e$.
P_k	Covariance matrix at k timestep.
$P_k^{(i)}$	Covariance matrix of each model at k timestep.
$P_{k-Ne}^{(i)}$	Covariance matrix of each model's starting state $k \geq N_e$.
$p_{lat,k}$	Lateral position at k timestep.
$p_{lon,k}$	Longitudinal position at k timestep.
p_{ref}	Lateral position of the lane being tracked.
T	Sampling time.
$t_{gap,k}$	Longitudinal time gap at k timestep.
$u_{lon,k}^{(AP)}$	Control inputs for the AP model.
$u_{lon,k}^{(DK)}$	Control inputs for the DK model.
$u_{lon,k}^{(LC)}$	Control inputs for the LC model.
$u_{lon,k}^{(VT)}$	Control inputs for the VT model.
$v_{lead,k}$	Longitudinal velocity of the leading vehicle.
$v_{ref,k}$	Velocity reference at k timestep.
$v_{x,k}$	Longitudinal velocity at k timestep.
$v_{y,k}$	Lateral velocity at k timestep.
$x_0^{(i)}$	Starting state of each model at $k < N_e$.
$x_{k-Ne}^{(i)}$	Starting state of each model at $k \geq N_e$.
$x_{lat,k}$	Lateral state at k timestep.
$x_{lead,k}$	Longitudinal state of the leading vehicle.
$x_{lon,k}$	Longitudinal state at k timestep.

I. Introduction

THE development of autonomous vehicles has significantly contributed to minimizing driver labor and reducing traffic accident rates. In a complex environment containing vehicles, pedestrians and obstacles, accurate estimation of the state information of surrounding vehicles is important for predicting the motions of these vehicles and thus more effectively supporting accurate trajectory planning for autonomous vehicles.

To accurately estimate the surrounding vehicle state information, many methods have been proposed. In [1], a front vehicle longitudinal acceleration estimator was proposed based on the Singer acceleration model, which uses the radar sensor to obtain the position and velocity of the front vehicle with good estimation results. Wu et al. [2] proposed an adaptive Kalman filter to estimate the acceleration of the front vehicle and used the estimated acceleration as a feedforward signal for the ego vehicle CACC controller in case of communication interruption, which achieved a good control effect. In [3], a multi-vehicle motion state estimation algorithm was proposed to accurately estimate the position and velocity information of multiple surrounding vehicles using Kalman filtering in combination with the Helly model and a map model. Kumar et al. [4] proposed an accurate fusion method to estimate the distance between an autonomous vehicle and other vehicles, objects, and signboards on the road. Sensor fusion was performed between a camera and LiDAR using a 3D marker based on the geometrical transformation and projection, and then a RefineDet detector was used to obtain the distance to the objects. In [5], a framework that uses four visual sensors for a full surround view of a vehicle was proposed for estimating the trajectories of surrounding vehicles in a highway environment. Trajectories are estimated using a vehicle detector together with a multiperspective optimized tracker in each view. The trajectories are transformed to a common ground plane, where they are associated between perspectives and analyzed to reveal tendencies around the ego vehicle.

Besides the use of radar sensors and cameras to obtain state information about the surrounding vehicles, with the development of communication technology, a variety of inter-vehicle communication and vehicle identification methods have emerged, such as which include V2V (Vehicle-to-vehicle), V2I (Vehicle-to-infrastructure), and Vehicle Recognition Technology [6]. In [7], a control framework adopting the on-board radar sensor and V2V communication has been proposed to fulfill the automated vehicle following in the longitudinal and lateral directions. Wherein, a path estimation method was proposed to calculate the trajectory of the preceding vehicle based on its historical movement data transmitted by V2V. In addition, Wang et al. [8] proposed an interactive multi-model event-triggered unscented Kalman filter (IMMETUKF) to estimate the state of the front vehicle using V2V communication technology. An event-triggered unscented Kalman filter (ETUKF) is first employed to estimate the front vehicle state to strike a balance between estimation accuracy and communication cost. To address the problem that single-model estimation methods for the front vehicle state are usually not applicable to multiple maneuvers of the vehicle, an interactive multi-model (IMM) approach is combined with ETUKF to form IMMETUKF to further improve the estimation accuracy and applicability. The IMM algorithm is a multi-model estimation algorithm that describes the behavioural characteristics of the estimated object by designing multiple models, each of which corresponds to one behavioural characteristic. The IMM algorithm estimates each model separately after the initial value interaction and fuses the estimates from each model using the updated model probabilities to finally obtain accurate fusion estimations. Applied to the automobile field, the estimation accuracy is improved by designing multiple models to match multiple maneuvers of the ego vehicle or surrounding vehicles.

Meanwhile, the multi-model estimation algorithms are constantly being updated and improved for further estimation accuracy. From the earliest static multimodel estimation algorithms [9], to later dynamic multimodel estimation algorithms [10], [11]. Xie et al. [12] proposed an interacting multiple model Kalman filter (IMM-KF) based on an adaptive transition probability, which eliminates the errors caused by fixed transition probability considering only prior information, improves the response speed of switching between models, and ensures the estimation accuracy. In [13], a multimodel particle filter was proposed to solve the joint traffic estimation and incident detection problem, and this filter was further extended to a multimodel particle smoother to improve the estimation accuracy when sensor data are limited.

In addition, the above mentioned multi-model estimation algorithm has a fixed number of models and estimators involved in the estimation in each iteration cycle, which not only increases the computational burden but also introduces models that do not match the current maneuver of the vehicle, thus leading to reduced estimation accuracy. To address this shortcoming, variable-structure multimodel estimation has received much attention. A variable-structure multimodel estimation algorithm abandons the assumption that the actual model space and the model set are the same and allows the model set to be adjusted in real time to follow the current external conditions and information. Existing methods for variable-structure multimodel estimation are divided into two main categories: (1) adaptive switching or selection in an overcomplete model set and (2) online real-time generation of new model sets based on target state estimation [14]. Li et al. [15] described the model group switching (MGS) method, which belongs to the first category and uses an approach of model activation and termination. The method is affected by the division and topology of the model set, which can reduce the estimation accuracy when the target maneuver changes too quickly. In addition, X.R. Li proposed the likely model set (LMS) [16] method, an algorithm that can reduce the computational burden while ensuring estimation accuracy, which also belongs to the first category. In [17], a variable-structure Gaussian mixture probability hypothesis density (GM-PHD) filter method based on the LMS algorithm was proposed. This method first determines the model set at each timestep using the LMS algorithm and then estimates it using the VSMM-GM-PHD algorithm. A modified likely-model set (MLMS) algorithm has also been proposed that combines the concepts of both the MGS and LMS estimators to achieve better tracking of quick changes [18]. The second class of variable-structure multimodel estimation methods includes the expected-mode augmentation (EMA) algorithm [19], [20] and the adaptive grid (AG) algorithm [21]. The former is an activation model algorithm that divides the model set into a base model set and an expected model set and ensures the estimation accuracy by activating the model that is closest to the target real model. The latter generates new models based on prior information of the target and, at each timestep, adaptively switches the center and distance of the grid to generate new models based on both prior and posterior information, thereby performing variable-structure multimodel estimation [22]. However, both fixed-structure multi-model estimation algorithms and variable-structure multi-model estimation algorithms only utilize the information from the last historical timestep and do not utilize sufficient historical information to further improve the accuracy, especially as the vehicle's traveling process is continuous.

For complex environments, Fang et al. [23] proposed a multimodal two-stage trajectory prediction framework for vehicle and pedestrian motion prediction. In the first stage, the trajectory set is generated in the form of hypothesis proposals, and in the second stage, the trajectory set is classified and refined to generate the final prediction, thereby achieving an excellent prediction effect. In [24], an IMM-KF algorithm was proposed to estimate the motion states of vehicles and predict their motion using model probability and vehicle kinematic models, both with good results.

To address the open questions, based on V2V communication technology, this paper designs multi-model estimation algorithms to estimate the state of the surrounding vehicles and predict the motion trajectories of the surrounding vehicles in a short period in the future. The main contributions of this paper are as follows:

- An IMM-MHE algorithm is proposed, enabling both multimodel estimation and full use of historical information. Compared with the traditional IMM algorithm, it has higher estimation accuracy and stronger anti-interference ability. It can also accurately estimate the state information of surrounding vehicles in a complex environment containing vehicles, pedestrians and obstacles.
- A VSIMM-MHE estimation framework is designed. First, a model classification method is designed to classify the models in the current model set into principle models, significant models and unlikely models. In each iteration, the models adjacent to the principle models are activated, the principle and significant models are retained, and some less likely models are eliminated. Time-domain adaptation is proposed to solve the problem that the fixed time domain of some models cannot be filled due to model activation and elimination. A new interaction method is proposed to solve the problem that models cannot interact because the starting timesteps of their time domains are different. The computational burden is reduced, and the final estimation error caused by the model not matching the current maneuver is also reduced.
- For a model set consisting of multiple intent-based motion models, a VSIMM-MHE algorithm is proposed. This algorithm introduces residual information into the model classification method, which reduces the dependence on the model probability accuracy, and thus reduces the influence of the priori transition probability matrix on the accuracy. Meanwhile, the model set of this variable structure estimation algorithm can be composed of different kinds of models for a wider range of applications. It can not only accurately estimate the state information of surrounding vehicles in complex environments but also effectively identify the current maneuvers of vehicles through a two-stage process, thereby effectively predicting the motion trajectories of surrounding vehicles. This provides strong informational support for the trajectory planning of autonomous vehicles.

The rest of this paper is structured as follows. In Section II, the motion models applied in this study to consider the presence of vehicles, obstacles and pedestrians are presented. In Section III, the IMM-MHE method is proposed. In Section IV, the VSIMM-MHE method is proposed. In Section V, simulation results are reported to verify the validity of the two proposed estimation methods. In addition, the principle of surrounding vehicle motion prediction is illustrated and simulation results are presented to demonstrate the effectiveness of prediction. Finally, conclusions are presented in Section VI.

II. Vehicle Models

To obtain information about the current states of the vehicles around an autonomous vehicle and from a basis for the subsequent motion prediction of the surrounding vehicles, several different intention-based motion models are utilized here. These models use the Earth coordinate system, and the longitudinal and lateral states are

$$x_{lon,k} = [p_{lon,k} \quad v_{x,k} \quad a_{x,k}]^T$$
$$x_{lat,k} = [p_{lat,k} \quad v_{y,k} \quad a_{y,k}]^T.$$

This section describes first the longitudinal and lateral motion models that match the different maneuvers of the vehicle and then the model sets that consist of these motion models. These motion models apply LQR in reverse, transforming the "reference inputs" into extended variables that are estimated in real time by subsequent estimation algorithms. The discrete kinematic formulations of these models are

$$a_{k+1} = a_k + u_k T$$
$$v_{k+1} = v_k + a_k T + \frac{1}{2}u_k T^2$$
$$p_{k+1} = p_k + v_k T + \frac{1}{2}a_k T^2 + \frac{1}{6}u_k T^3.$$

Translated into the form of the state space equation as

$$x_{k+1} = \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} T^6/3 \\ T^2/2 \\ T \end{bmatrix} u_k.$$

More specific details and explanations about these models and model sets can be found in our previous work [25].

A. Longitudinal Motion Models

For longitudinal motion, the distance keeping (DK) model, the velocity tracking (VT) model and the avoidance parking (AP) model are used to obtain the state information and intentions of the surrounding vehicles.

1) Velocity Tracking: The VT model represents the driving maneuver of a vehicle following the longitudinal velocity of the vehicle or pedestrian in front of it, and its discrete-time equation and control input are

$$x_{lon,k+1} = \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} x_{lon,k} + \begin{bmatrix} 0 \\ T^2/2 \\ T \end{bmatrix} u_{lon,k}^{(VT)}$$
$$u_{lon,k}^{(VT)} = -K_{LQR}^{(VT)}(v_{x,k} - v_{ref,k}).$$

It is worth noting that since only $v_{x,k}$ and $v_{ref,k}$ were utilized in the calculation of the control inputs and no positional variables were utilized, $T^3/6$ was replaced by 0. With the unknown estimation variable $v_{ref,k}$ as the extended state variable, the VT model becomes

$$\begin{bmatrix} x_{lon,k+1} \\ v_{ref,k+1} \end{bmatrix} = F_{lon}^{(VT)}(K_{LQR}^{(VT)}) \begin{bmatrix} x_{lon,k} \\ v_{ref,k} \end{bmatrix} + \omega_k^{(VT)} \tag{1}$$

where $F_{lon}^{(VT)}(K_{LQR}^{(VT)})$ is the state transfer matrix of the VT model and $\omega_k^{(VT)}$ allows for tracking variations in $v_{ref,k}$.

2) Distance Keeping: The DK model represents the cruise control maneuver of maintaining a specified distance from the vehicle in front, and its discrete-time equation and control input are

$$\tilde{x}_{lon,k+1} = \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} \tilde{x}_{lon,k} + \begin{bmatrix} T^3/6 \\ T^2/2 \\ T \end{bmatrix} u_{lon,k}^{(DK)}$$
$$\tilde{x}_{lon,k} = x_{lon,k} - x_{lead,k}$$
$$u_{lon,k}^{(DK)} = -K_{LQR}^{(DK)}([p_{lon,k} - p_{lead,k}] - [-v_{lead,k}t_{gap,k}]).$$

Bringing the control input formula into the discrete time equation and taking the unknown state variable $t_{gap,k}$ as an extended state variable, the DK model becomes

$$\begin{bmatrix} x_{lon,k+1} \\ t_{gap,k+1} \end{bmatrix} = F_{lon,k}^{(DK)}(K_{LQR}^{(DK)}, \chi_k) \begin{bmatrix} x_{lon,k} \\ t_{gap,k} \end{bmatrix} + E_{lon,k}^{(DK)}(K_{LQR}^{(DK)}, \chi_{k:k+1}) + \omega_k^{(DK)} \tag{2}$$

where $E_{lon,k}^{(DK)}(K_{LQR}^{(DK)}, \chi_{k:k+1})$ is the external input, χ_k denotes the state information of the leading vehicle, and $\omega_k^{(DK)}$ allows for tracking variations in $t_{gap,k}$.

3) Avoidance Parking: The AP model represents the driving maneuver of parking to avoid a pedestrian or obstacle, and its discrete-time equation and control input are

$$\tilde{x}_{lon,k+1} = \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} \tilde{x}_{lon,k} + \begin{bmatrix} T^3/6 \\ T^2/2 \\ T \end{bmatrix} u_{lon,k}^{(AP)}$$
$$\tilde{x}_{lon,k} = x_{lon,k} - x_{lead,k}$$
$$u_{lon,k}^{(AP)} = -K_{LQR}^{(AP)}(\tilde{x}_{lon,k} - [d_{pre,k} \quad 0 \quad 0]^T).$$

By treating $d_{pre,k}$ as an extended state variable, the AP model becomes

$$\begin{bmatrix} x_{lon,k+1} \\ d_{pre,k+1} \end{bmatrix} = F_{lon,k}^{(AP)}(K_{LQR}^{(AP)}, \chi_k) \begin{bmatrix} x_{lon,k} \\ d_{pre,k} \end{bmatrix} + E_{lon,k}^{(AP)}(K_{LQR}^{(AP)}, \chi_{k:k+1}) + \omega_k^{(AP)} \tag{3}$$

where $\omega_k^{(AP)}$ allows for tracking variations in $d_{pre,k}$.

The Q matrix of LQR is a semi-positive definite matrix whose elements on the diagonal correspond to the relative importance of the state variables. Increasing the weight of a state variable in the Q matrix will result in the controller placing more emphasis on the regulation of this state variable. However, if Q is set too large, it may make the system too sensitive and react strongly to noise. Setting Q too small may result in a slow or insufficiently stable system state response. The R matrix of LQR is a positive definite matrix used to adjust the cost of the control inputs. Increasing R reduces the controller's dependence on the control inputs, decreasing R makes the controller more dependent on the control inputs to regulate the state, which may result in a faster response.

Since the LQR is applied in reverse, the measured vehicle signals are used to estimate the control variables, which in turn solve for the "reference inputs", i.e. the extended variables in the paper. The subsequent proposed estimation algorithms also have a filtering effect when estimating the control variables, which can effectively deal with the measurement noise and weaken the jerk nature of the control variables.

B. Lateral Motion Models

There are three options available to a moving vehicle when faced with the possibility of a lane change: stay in the current lane, switch lanes to the left, or switch lanes to the right. These three lane-changing (LC) maneuvers correspond to three models, represented by $m^{(LC1)}$, $m^{(LC2)}$, and $m^{(LC3)}$, respectively.

1) Lane Changing: The LC models represent lateral driving maneuvers that track the lateral position of a lane. Their discrete-time equations and control inputs are

$$x_{lat,k+1} = \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} x_{lat,k} + \begin{bmatrix} T^3/6 \\ T^2/2 \\ T \end{bmatrix} u_{lat,k}^{(LC)}$$

$$u_{lat,k}^{(LC)} = -K_{LQR}^{(LC)}(x_{lat,k} - [p_{ref} \quad 0 \quad 0]^T).$$

Finally, the LC models are written as

$$x_{lat,k+1} = F_{lat,k}^{(LC)}(K_{LQR}^{(LC)})x_{lat,k} + E_{lat,k}^{(LC)}(K_{LQR}^{(LC)}, p_{ref}) + \omega_k^{(LC)} \tag{4}$$

where $\omega_k^{(LC)}$ allows for tracking variations in the reference lane's lateral position.

C. Model Sets

Finally, the above longitudinal and lateral motion models are combined to produce two model sets. The model set $\mathcal{M}1 = \{VT - LC1, VT - LC2, VT - LC3, DK - LC1, DK - LC2, DK - LC3\}$ obtained by combining the VT, DK, and LC models is used for situations in which the surrounding environment is full of vehicles. The model set $\mathcal{M}2 = \{VT - LC1, VT - LC2, VT - LC3, AP - LC1, AP - LC2, AP - LC3\}$ obtained by combining the VT, AP and LC models is used for situations in which there are pedestrians or obstacles in the surrounding environment. The difference between these two model sets is the DK model and the AP model. The DK model represents the maneuver of maintaining a constant distance from the vehicle in front, which is object-oriented to the vehicle, while the AP model represents the maneuver of stopping to avoid pedestrians or obstacles, which is object-oriented to the pedestrians and obstacles. While the vehicle is traveling, there is no guarantee that the surroundings will always have vehicles, pedestrians, and obstacles at the same time, so two model sets are needed. Detection of vehicles, pedestrians, and obstacles in the surrounding environment by LiDAR, which in turn selects the corresponding model set. To describe each model in each model set more clearly, the discrete-time equation and the observation equation are given by

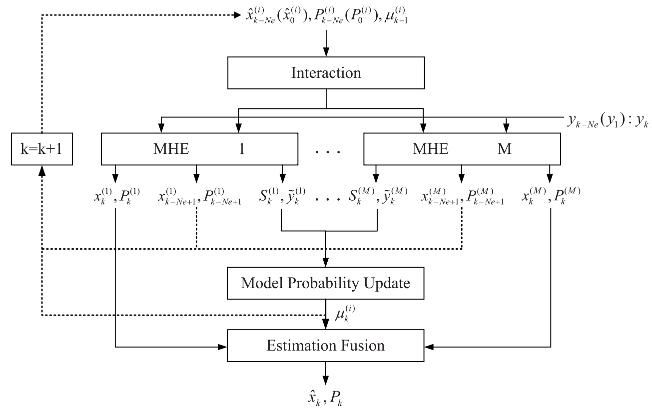


Fig. 1. Schematic of the IMM-MHE algorithm.

$$x_{k+1}^{(i)} = F_i x_k^{(i)} + E_k^{(i)} + \omega_k^{(i)}$$
$$y_k = H_i x_k^{(i)} + \upsilon_k^{(i)}.$$

The superscript (i) is used to denote the different models in the model set, and the subscript k denotes different timesteps. F_i and $E_k^{(i)}$ consist of a longitudinal motion model and a lateral motion model:

$$F_i = \begin{bmatrix} F_{lon,k} & 0 \\ 0 & F_{lat,k} \end{bmatrix}, \quad E_k = \begin{bmatrix} E_{lon,k} \\ E_{lat,k} \end{bmatrix}.$$

The state variable is $x = [p_{lon,k}, v_{x,k}, a_{x,k}, p_{lat,k}, v_{y,k}, a_{y,k}, v_{ref,k}/t_{gap,k}/d_{pre,k}]^T$, the measurement variable is $y = [p_{lon,k}, v_{x,k}, a_{x,k}, p_{lat,k}, v_{y,k}, a_{y,k}]^T$, and the external input is $E_k = [p_{lead,k}, v_{lead,k}, a_{lead,k}]^T$, which is acquired by on-board sensors and V2V communication technology. It is worth stating that all the above proposed vehicle motion models are point mass models. Point mass models are often used to estimate vehicle state information as well as to predict future vehicle trajectories [24]. Meanwhile, the longitudinal and lateral motions of the vehicle are coupled during travelling, but the coupling relationship between the longitudinal motion and lateral motion models is not considered. This work aims to achieve state estimation and motion prediction of surrounding vehicles using relatively simple models and to improve the accuracy of estimation and prediction by designing new multi-model estimation algorithms, thereby reducing the difficulty of establishing higher precision models.

III. State Estimation Based on IMM-MHE

To match the multiple maneuvers of vehicles and to make full use of the vehicle history information, a new IMM-MHE method is proposed. This method uses MHE to estimate each model in the model set to match multiple maneuvers during vehicle driving while following the interaction, probability update and combination steps of the IMM algorithm, which is a multimodel estimation method.

The algorithm procedures are described below, divided into Steps 1–4. The algorithm outputs the fused state estimate $\hat{x}_k$ and covariance matrix P_k at the current timestep, and also outputs each model's state estimate $\hat{x}_k^{(i)}$, covariance matrix $P_k^{(i)}$

and the model probability $\mu_k^{(i)}$ for the next iteration. As shown in Figure 1, the input values are subjected to interaction and the values after interaction are provided to the corresponding models in the model set. Each model in the model set is estimated using the formulas in Step 2. Subsequently, the model probabilities are updated using Eqs. (22) and (23). Finally, the states of each model are fused in accordance with the model probabilities to obtain an estimate at the current time. The algorithm is described as follows:

Step 1a (Interaction at $k < N_e$):

When the time k is smaller than the time-domain length N_e, equivalent to full information estimation (FIE), the interaction step is as follows:

$$c^{(i)} = \sum_{j=1}^{M} \pi_{ji} \mu_{k-1}^{(j)} \tag{5}$$

$$\mu_{k-1}^{(j|i)} = \frac{\pi_{ji} \mu_{k-1}^{(j)}}{c^{(i)}} \tag{6}$$

$$\overline{x}_0^{(i)} = \sum_{j=1}^{M} \mu_{k-1}^{(j|i)} \hat{x}_0^{(j)} \tag{7}$$

$$\overline{P}_0^{(i)} = \sum_{j=1}^{M} \mu_{k-1}^{(j|i)} (P_0^{(j)} + X_0^{(i,j)}) \tag{8}$$

where $X_k^{(i,j)} = (\overline{x}_k^{(i)} - \hat{x}_k^{(j)})(\overline{x}_k^{(i)} - \hat{x}_k^{(j)})^T$. Since the MHE is not rolling in this case, a priori estimate and covariance of each model's starting state are fixed, denoted by $\hat{x}_0^{(i)}$ and $P_0^{(i)}$, respectively. An overline is used to indicate the variables that are obtained after interaction.

Step 1b (Interaction at $k \geq N_e$):

When the time k is larger than the time-domain length N_e, the interaction step is as follows:

$$c^{(i)} = \sum_{j=1}^{M} \pi_{ji} \mu_{k-1}^{(j)} \tag{9}$$

$$\mu_{k-1}^{(j|i)} = \frac{\pi_{ji} \mu_{k-1}^{(j)}}{c^{(i)}} \tag{10}$$

$$\overline{x}_{k-Ne}^{(i)} = \sum_{j=1}^{M} \mu_{k-1}^{(j|i)} \hat{x}_{k-Ne}^{(j)} \tag{11}$$

$$\overline{P}_{k-Ne}^{(i)} = \sum_{j=1}^{M} \mu_{k-1}^{(j|i)} (P_{k-Ne}^{(j)} + X_{k-Ne}^{(i,j)}) \tag{12}$$

Since the MHE is rolling in this case, a priori estimate and covariance of each model's starting state are changed, denoted by $\hat{x}_{k-Ne}^{(i)}$ and $P_{k-Ne}^{(i)}$, respectively.

Step 2a (Mode-matched MHE at $k < N_e$):

When the time k is less than the time-domain length Ne, the data from all previous periods are used. By solving the following optimization problem:

$$\begin{aligned} &\min_{x_0^{(i)}, \{\omega_k^{(i)}\}_{k=0}^{k-1}} \Phi_k(x_0^{(i)}, \{\omega_k^{(i)}\}) \\ &\Phi_k(x_0^{(i)}, \{\omega_k^{(i)}\}) = \sum_{j=0}^{k} ||\upsilon_j^{(i)}||_{R^{-1}}^2 + \sum_{j=0}^{k-1} ||\omega_j^{(i)}||_{Q^{-1}}^2 \\ &\quad + ||x_0^{(i)} - \overline{x}_0^{(i)}||_{\overline{P}_0^{(i)-1}}^2 \\ &s.t. \quad x_{k+1}^{(i)} = F_i x_k^{(i)} + E_k^{(i)} + \omega_k^{(i)} \\ &\qquad y_k = H_i x_k^{(i)} + \upsilon_k^{(i)} \end{aligned} \tag{13}$$

we can obtain $(x_0^{(i)}, \{\omega_k^{(i)}\})$. The first part of the cost function is the measurement noise term of the system, the second part is the process noise term, and the third part is the starting state estimation term. Where the measurement information is introduced in the first part to correct the estimation error and the optimization variable $(x_0^{(i)}, \{\omega_k^{(i)}\})$ is calculated by minimizing the cost function. Q and R are coefficient matrices, the former expressing confidence in the model and the latter expressing confidence in the measurements. From $(x_0^{(i)}, \{\omega_k^{(i)}\})$, the following can be calculated:

$$\hat{x}_k^{(i)} = (F_i)^k x_0^{(i)} + \sum_{j=0}^{k-1} (F_i)^{k-j-1} E_j^{(i)} + \sum_{j=0}^{k-1} (F_i)^{k-j-1} \omega_j^{(i)} \tag{14}$$

$$P_1^{(i)} = Q + F_i(\overline{P}_0^{(i)} - \overline{P}_0^{(i)} H_i^T (R + H_i \overline{P}_0^{(i)} H_i^T)^{-1} H_i \overline{P}_0^{(i)}) F_i^T \tag{15}$$

The covariance matrices $P_{k-1}^{(i)}$ and $P_k^{(i)}$ can be calculated through multiple iterations of (15). In addition, the covariance $S_k^{(i)}$ and residual $\tilde{y}_k^{(i)}$ can be calculated from the following equations:

$$S_k^{(i)} = H_i (F_i P_{k-1}^{(i)} F_i^T + Q) H_i^T + R \tag{16}$$

$$\tilde{y}_k^{(i)} = y_k - H_i x_k^{(i)} \tag{17}$$

Step 2b (Mode-matched MHE at $k \geq N_e$):

When the time k is larger than the time-domain length N_e, by solving the following optimization problem:

$$\begin{aligned} &\min_{x_{k-N_e}^{(i)}, \{\omega_k^{(i)}\}_{k-N_e}^{k-1}} \Phi_k(x_{k-N_e}^{(i)}, \{\omega_k^{(i)}\}) \\ &\Phi_k(x_{k-N_e}^{(i)}, \{\omega_k^{(i)}\}) = \sum_{j=k-N_e}^{k} ||\upsilon_j^{(i)}||_{R^{-1}}^2 \\ &\quad + \sum_{j=k-N_e}^{k-1} ||\omega_j^{(i)}||_{Q^{-1}}^2 \\ &\quad + ||x_{k-N_e}^{(i)} - \overline{x}_{k-N_e}^{(i)}||_{\overline{P}_{k-N_e}^{(i)-1}}^2 \\ &s.t. \quad x_{k+1}^{(i)} = F_i x_k^{(i)} + E_k^{(i)} + \omega_k^{(i)} \\ &\qquad y_k = H_i x_k^{(i)} + \upsilon_k^{(i)} \end{aligned} \tag{18}$$

we can obtain $(x_{k-N_e}^{(i)}, \{\omega_k^{(i)}\})$. The following can be calculated:

$$\hat{x}_{k-Ne+1}^{(i)} = F_i x_{k-N_e}^{(i)} + E_{k-Ne}^{(i)} + \omega_{k-Ne}^{(i)} \tag{19}$$

$$\hat{x}_k^{(i)} = (F_i)^{N_e} x_{k-N_e}^{(i)} + \sum_{j=k-N_e}^{k-1} (F_i)^{k-j-1} E_j^{(i)} + \sum_{j=k-N_e}^{k-1} (F_i)^{k-j-1} \omega_j^{(i)} \tag{20}$$

$$P_{k-N_e+1}^{(i)} = Q + F_i(\overline{P}_{k-N_e}^{(i)} - \overline{P}_{k-N_e}^{(i)} H_i^T (R + H_i \overline{P}_{k-N_e}^{(i)} H_i^T)^{-1} H_i \overline{P}_{k-N_e}^{(i)}) F_i^T \tag{21}$$

Similarly, the covariance matrices $P_{k-1}^{(i)}$ and $P_k^{(i)}$ can be calculated though multiple iterations of (21), and the covariance $S_k^{(i)}$ and residual $\tilde{y}_k^{(i)}$ are also obtained from (16) and (17), respectively.

Step 3 (Model probability update):

For each model, the model probabilities $\mu_k^{(i)}$ can be updated as follows:

$$L_k^{(i)} = \frac{exp(-\frac{1}{2}\tilde{y}_k^{(i)^T} S_k^{(i)^{-1}} \tilde{y}_k^{(i)})}{|2\pi S_k^{(i)}|^{\frac{1}{2}}} \tag{22}$$

$$\mu_k^{(i)} = \frac{c^{(i)} L_k^{(i)}}{\sum_{j=1}^{M} c^{(j)} L_k^{(j)}} \tag{23}$$

Step 4 (Estimate fusion):

The state estimate $\hat{x}_k$ and its covariance P_k, which combine all model-based state estimates and associated covariance matrices, are

$$\hat{x}_k = \sum_{i=1}^{M} \mu_k^{(i)} \hat{x}_k^{(i)} \tag{24}$$

$$P_k = \sum_{i=1}^{M} \mu_k^{(i)} [P_k^{(i)} + (\hat{x}_k - x_k^{(i)})(\hat{x}_k - x_k^{(i)})^T] \tag{25}$$

The pseudocode of the IMM-MHE algorithm is summarized in Algorithm 1. The IMM-MHE algorithm can match multiple vehicle maneuvers while fully utilizing historical information, thus enabling accurate estimation of the state information of surrounding vehicles. In addition, it is worth stating that the increase of the time-domain length N_e can improve the estimation accuracy, but it will increase the computational burden, so the estimation accuracy and computational burden need to be taken into account when selecting N_e.

Algorithm 1 IMM-MHE: Recursive, Timestep k

```
Input: x̂_0^(i), P_0^(i), μ_{k-1}^(i) or x̂_{k-Ne}^(i), P_{k-Ne}^(i), μ_{k-1}^(i)
if k < Ne then
    Measurement: y_1 : y_k ← system
    Interaction: x̄_0^(i), P̄_0^(i) ← (5)(6)(7)(8)
    for i = 1 : M do
        Estimation: x_k^(i) ← (13)(14)
        Compute: P_k^(i), S_k^(i), ỹ_k^(i) ← (15)(16)(17)
    end
else
    Measurement: y_{k-Ne} : y_k ← system
    Interaction: x̄_{k-Ne}^(i), P̄_{k-Ne}^(i) ← (9)(10)(11)(12)
    for i = 1 : M do
        Estimation: x̂_{k-Ne+1}^(i), x̂_k^(i) ← (18)(19)(20)
        Compute: P_{k-Ne+1}^(i), P_k^(i), S_k^(i), ỹ_k^(i) ← (21)(16)(17)
    end
end
Model probability: μ_k^(i) ← (22)(23)
Estimate fusion: x̂_k, P_k ← (24)(25)
Output: x̂_k, P_k, μ_k^(i) or x̂_k, P_k, x̂_{k-Ne+1}^(i), P_{k-Ne+1}^(i), μ_k^(i)
```

IV. Moving Horizon Estimation With Variable Structure Interacting Multiple Model

To reduce the computational burden and the final fusion-based estimation error from models that do not match the current maneuver, a VSIMM-MHE framework is proposed. In this framework, the models in the model set are classified into principle models, significant models and unlikely models using a new model classification method. The models adjacent to the principle models are activated, the principle and significant models are retained, and some less likely models are eliminated. The adjacent models mean that the model transfer probability between two models is greater than 0, i.e., $\pi_{ij} > 0$. To solve the problem that the fixed time domain of some models cannot be filled due to model activation and elimination, time-domain adaptation is proposed. A new interaction method is also proposed to overcome the problem that models cannot interact because the starting timesteps of their time domains are different. Based on the background of estimation and prediction of surrounding vehicles, a VSIMM-MHE algorithm is proposed. This algorithm introduces residual information into the model classification method, reducing the dependence on the accuracy of the model probabilities, and its model set consists of different kinds of intent-based motion models. It can not only accurately estimate the state information of surrounding vehicles in a complex environment, but also identify the model that best matches the current maneuver and effectively predict the motion trajectories of surrounding vehicles through model probabilities.

In the rest of this section, the model classification method, time-domain adaptation and interaction method for VSIMM-MHE are sequentially described in detail. The complete VSIMM-MHE algorithm is then illustrated, and the effects of the thresholds t_1 and t_2 on the estimation performance are analyzed. Finally, the kinds of models to which the algorithm applies are described.

A. Model Classification

To reduce the dependence on the accuracy of the model probabilities, a new model classification method is proposed. Specifically, the model classification method used in the LMS algorithm is redesigned. Run IMM-MHE[$\mathbb{M}_k, \mathbb{M}_{k-1}$] to obtain the model probability $\mu_k^{(i)}$ for each model in the model set $\mathbb{M}_k$, and then the residual information corresponding to each model at the current timestep is introduced to obtain the corresponding variable $\lambda_k^{(i)}$, as expressed in the following equation:

$$\lambda_k^{(i)} = (\mu_k^{(i)})^{-1} ||\tilde{y}_k^{(i)}||_2 \tag{26}$$

where $\tilde{y}_k^{(i)}$ is obtained from (17). IMM-MHE[$\mathbb{M}_k, \mathbb{M}_{k-1}$] denotes IMM-MHE from $\mathbb{M}_{k-1}$ to $\mathbb{M}_k$, and $\mathbb{M}_{k-1}$ is the set of models output at the previous timestep, which is used to identify the models participating in IMM-MHE, details can be found in Section III. It is worth noting that both the proposed IMM-MHE algorithm and the traditional IMM

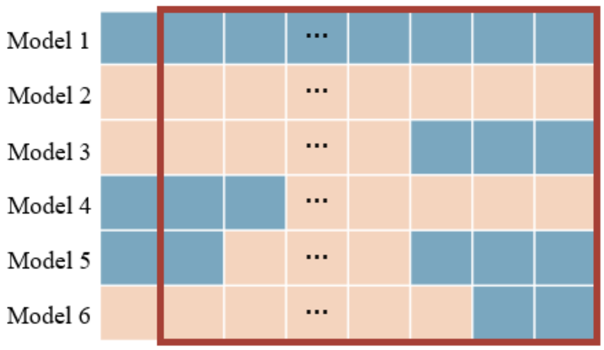


Fig. 2. The distribution of information in the time domain at k timesteps. The brown box indicates the time domain of MHE. The horizontal coordinates are timesteps, the rightmost timestep in the time domain corresponds to k timestep, the time decreases to the left, and the leftmost timestep in the time domain indicates $k - N_e$ timestep. Model 1-6 correspond to different models in model sets $\mathcal{M}1$ and $\mathcal{M}2$, the orange grid represents the absence of information at that timestep and the blue grid represents the presence of information at that timestep. Grids do not have numerical significance.

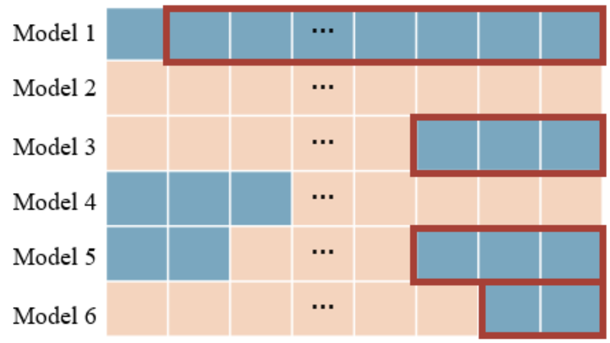


Fig. 3. The time domain of each model involved in the estimation at k timesteps. The brown box indicates the time domain of MHE. The horizontal coordinates are timesteps, Model 1-6 correspond to different models in model sets $\mathcal{M}1$ and $\mathcal{M}2$. The orange grid represents the absence of information at that timestep, and the blue grid represents the presence of information at that timestep.

algorithm perform soft switching based on the state transfer matrix π, and the response to the model jumps of the system takes some time, so the model probabilities computed based on the priori state transfer matrix have a certain delay with the real models jumps. To classify the model more accurately, Eq. (26) attenuates the negative effect of the model probability delay by introducing residual information.

By comparing the magnitudes of the $\lambda_k^{(i)}$ variables against two thresholds t_1 and t_2, the models in model set $\mathbb{M}_k$ can be classified into three types: principle models ($\lambda_k^{(i)} < t_2$), significant models ($t_2 \leq \lambda_k^{(i)} \leq t_1$), and unlikely models ($\lambda_k^{(i)} > t_1$). The smaller $\lambda_k^{(i)}$ is, the more likely it is that the model corresponds to the current maneuver. Meanwhile, due to the introduction of the residual information, the range of values of the thresholds t_1 and t_2 changes from the interval $[0, 1]$ to $[0, \infty)$.

B. Time-Domain Adaptation

In the IMM-MHE algorithm, estimation is performed using information from multiple historical timesteps; however, because of the variable structure, the problem may arise that no information is available at certain timesteps in the time domain. As shown in Figure 2, models that have not previously been involved in estimation in the time domain, such as Model 6, may be activated because they are adjacent to the principle models; for such models, only historical information from the previous timestep exists. A case may also arise in which the model has been activated before, but the information present still does not fill the entire time domain, for example, Model 3. In addition, there is the case of models that participate in estimation at the start of the time domain, are eliminated at some later timestep by becoming classified as unlikely models, but are eventually reactivated to participate in estimation again due to their adjacency to the principle models, such as model 5. The IMM-MHE algorithm fails to achieve estimation in all of the above three cases.

To solve the above problem, a time-domain adaptation method is proposed. For all models in the model set, a fixed-length time domain is first specified, and its length is denoted by ${N_e}^{max}$. Then, the last timestep with nonexistent information is found by searching in the direction from the beginning to the end of the fixed time domain, and only the historical information from the subsequent timestep $k - L_t$ to the current timestep k is used as the new time domain of this model, as shown in Figure 3. If no information exists at the last timestep in the whole time domain, the model is not involved in the estimation process. It is worth clarifying that for the case of Model 5, the historical information at the beginning of the fixed time domain is discarded and only the historical information close to the current timestep is kept. Therefore, the length of the time domain in each MHE in the VSIMM-MHE algorithm is variable. The details of the time-domain adaptation process are described in Algorithm 2.

C. Interaction Method for VSIMM-MHE

Since models are activated and eliminated, the time domain lengths of the models involved in the estimation at the current timestep are different, which results in their starting timesteps being different. For example, the starting timestep of model 1 in Figure 3 is the $k-N_e$ timestep, while the starting timestep of the time domain of model 6 is $k-1$. Therefore it is not possible for these models interact as they do in the traditional IMM algorithm. Therefore, we propose a new interaction method to meet the needs of the VSIMM-MHE algorithm. In this method, the interaction process between models is completed based on the models' time-domain lengths. As shown in Figure 4, models with the same time-domain length need to interact, and models with different time-domain lengths do not need to interact. If there is only one model with a certain time-domain length in model set $\mathbb{M}_k$, then a priori estimate and covariance of its starting state are directly output as the value after interaction, i.e., $\overline{x}_{k-Ne}^{(i)} = \hat{x}_{k-N_e}^{(i)}$, $\overline{P}_{k-N_e}^{(i)} = P_{k-N_e}^{(i)}$. Finally, all values obtained after interaction are given to the corresponding MHE estimators sequentially. The details are described in Algorithm 3. Because of the application of time-domain adaptation, there is no process with $k < N_e$. Therefore, only the case in which the time domain is rolling at the current timestep needs to be considered. It is worth noting that Figures 3 and 4

Algorithm 2 Time-Domain Adaptation: Recursive, Timestep k

Input: Fixed-length time domain for all models
for Model $i = 1 : M$ do
  if $k < N_e{}^{max}$
    for $j = 1 : k$ do
$$f_j = \begin{cases} 1, & \text{information exists at timestep } j \\ 0, & \text{no information exists at timestep } j \end{cases}$$
    end
  else
    for $j = k - N_e{}^{max} : k$ do
$$f_j = \begin{cases} 1, & \text{information exists at timestep } j \\ 0, & \text{no information exists at timestep } j \end{cases}$$
    end
  end
  Obtain the location L_0 of the last 0 in the queue
  if $L_0 \geq N_e{}^{max}$
    Time-domain length: $N_e^{(i)} = 0$
  else
    Compute: $L_t = L_0 + 1$
    Time-domain length: $N_e^{(i)} = \sum_{j=k-L_t}^{k} f_j$
  end
end
output: Time domain containing information about $N_e^{(i)}$ historical timesteps, and $N_e^{(i)}$

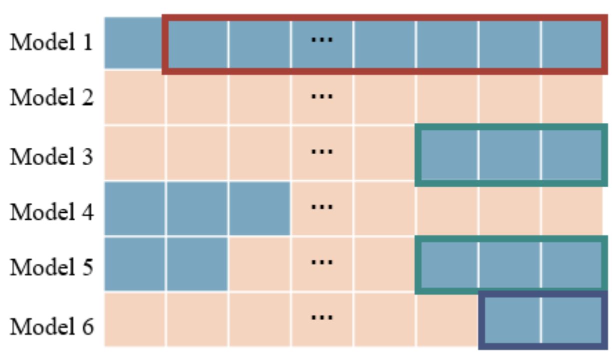


Fig. 4. The interaction schematic of the VSIMM-MHE algorithm at k timestep. The horizontal coordinates are timesteps, and each color represents a time domain length. Models with the same time domain length need to interact with each other, and models with different time domain lengths do not need to interact.

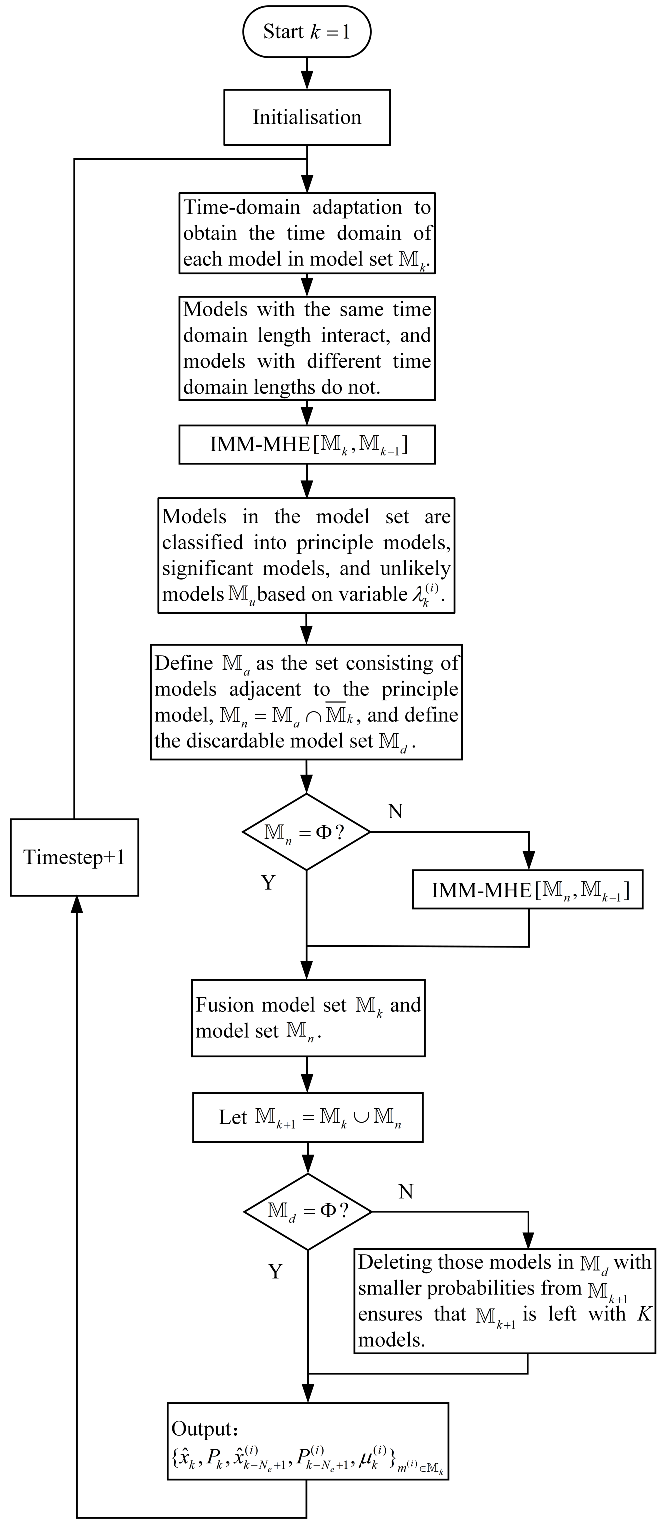


Fig. 5. Flowchart of VSIMM-MHE algorithm.

illustrates the time-domain adaptation and interaction of the VSIMM-MHE algorithm, respectively, where the estimated object is the "ego vehicle" of the subsequent simulation part.

Notably, in the VSIMM-MHE algorithm, if there are no fewer than 2 activated models, interactive multimodel estimation is required for these models. All these models use only historical information from the previous timestep and have the same starting timestep, so the IMM-MHE algorithm can be used directly.

The detailed VSIMM-MHE algorithm with AND logic is summarized in Table I. It is worth clarifying that all three algorithms discussed above are part of the VSIMM-MHE algorithm. In each iteration, Algorithm 2 is run first to obtain the time domain and the time-domain length $N_e^{(i)}$ of each model. Then, IMM-MHE$[\mathbb{M}_k, \mathbb{M}_{k-1}]$ is run to obtain $\mu_k^{(i)}$ and $\tilde{y}_k$, where IMM-MHE$[\mathbb{M}_k, \mathbb{M}_{k-1}]$ is the concrete representation of Algorithm 1. However, the interaction step in Algorithm 1 is replaced by Algorithm 3. The flowchart of the VSIMM-MHE algorithm is shown in Figure 5.

Eliminating the unlikely models refers to not allowing these models to participate in the calculation of estimated values in the next iteration cycle. These unlikely models are activated

TABLE I

ONE CYCLE OF THE VSIMM-MHE ALGORITHM WITH AND LOGIC

S1 Time $k = k + 1$. First, run Algorithm 2 to obtain the time domain and $N_e^{(i)}$. Subsequently, run IMM-MHE$[\mathbb{M}_k, \mathbb{M}_{k-1}]$ to obtain $\mu_k^{(i)}$ and $\tilde{y}_k^{(i)}$, where Equations (9)(10)(11)(12) in the IMM-MHE$[\mathbb{M}_k, \mathbb{M}_{k-1}]$ are replaced by Algorithm 3.
S2 Obtain $\lambda_k^{(i)}$ via Equation (26).
S3 Classify the models in $\mathbb{M}_k$ into principle models ($\lambda_k^{(i)} < t_2$), significant models ($t_2 \le \lambda_k^{(i)} \le t_1$), and unlikely models ($\lambda_k^{(i)} > t_1$) based on $\lambda_k^{(i)}$. Let the model set consisting of unlikely models be denoted by $\mathbb{M}_u$. If $\mathbb{M}_u = \varnothing$ and there is no principle model, output $\{\hat{x}_k, P_k, \hat{x}_{k-Ne+1}^{(i)}, P_{k-Ne+1}^{(i)}, \mu_k^{(i)}\}_{m^{(i)} \in \mathbb{M}_k}$ directly. Let $\mathbb{M}_{k+1} = \mathbb{M}_k$, then return to S1. Otherwise, go to S4.
S4 Let $\mathbb{M}_a$ be defined as the set of all models adjacent to the principle models. If no principle model exists, set this model set to $\mathbb{M}_a = \varnothing$ and go to S5. Otherwise, obtain the model set $\mathbb{M}_n = \mathbb{M}_a \cap \overline{\mathbb{M}}_k$, with $\overline{\mathbb{M}}_k$ denoting the complement of $\mathbb{M}_k$. Then run IMM-MHE$[\mathbb{M}_n, \mathbb{M}_{k-1}]$ for each model in $\mathbb{M}_n$ to obtain the state estimates, covariance and model probability: $\{\hat{x}_k^{(i)}, P_k^{(i)}, \mu_k^{(i)}\}_{m^{(i)} \in \mathbb{M}_n}$. Subsequently, fuse the model sets $\mathbb{M}_k$ and $\mathbb{M}_n$ to obtain the global fused state estimates, covariance matrix, starting state estimates and covariances for each model at the next timestep, as well as model probabilities: $\{\hat{x}_k, P_k, \hat{x}_{k-N_e+1}^{(i)}, P_{k-N_e+1}^{(i)}, \mu_k^{(i)}\}_{m^{(i)} \in (\mathbb{M}_k \cup \mathbb{M}_n)}$. In addition, update $\mathbb{M}_k = \mathbb{M}_k \cup \mathbb{M}_n$.
S5 Output $\{\hat{x}_k, P_k, \hat{x}_{k-N_e+1}^{(i)}, P_{k-N_e+1}^{(i)}, \mu_k^{(i)}\}_{m^{(i)} \in \mathbb{M}_k}$.
S6 Define $\mathbb{M}_d$ as the set of discardable models, which are the unlikely models not adjacent to any principle model. If $\mathbb{M}_d = \varnothing$, then return to S1. Otherwise, go to S7.
S7 Let $\mathbb{M}_{k+1} = \mathbb{M}_k - \mathbb{M}_m$, where $\mathbb{M}_m \subset \mathbb{M}_d$ is the set of models with smaller probabilities in $\mathbb{M}_d$, ensuring that $\mathbb{M}_{k+1}$ has at least K models remaining. Otherwise, return S1.

Algorithm 3 Interaction Method for VSIMM-MHE: Recursive, Timestep k

```
Input: N_e^(i), π, x̂^(i)_{k−N_e}, P^(i)_{k−N_e}, μ^(i)_{k−1}
Find the maximum value N_e^m from N_e^(i)
for N_e = 1 : N_e^m do
  Calculate N_um and find corresponding models L_oc
  if N_um == 1
      x̄^(L_oc)_{k−N_e} = x̂^(L_oc)_{k−N_e}, P̄^(L_oc)_{k−N_e} = P^(L_oc)_{k−N_e}
  elseif N_um == 0
      Continue
  else
      Model probability normalization ← μ^(L_oc)_{k−1}
      Obtain transition probability: π′ = π(L_oc, L_oc)
      Interaction: x̄^(L_oc)_{k−Ne}, P̄^(L_oc)_{k−N_e} ← (9)(10)(11)(12)
  end
end
Output: x̄^(i)_{k−Ne}, P̄^(i)_{k−N_e}
```

again based on whether they are adjacent to the principle models. These unlikely models are activated when the models adjacent to these unlikely models are classified as principle models in a certain iteration, corresponding to S4 in Table I. At the same time, these activated unlikely models participate in the estimation using only historical information from the previous timestep and are subsequently reclassified.

The threshold values t_1 and t_2 are important factors in the algorithm and have a significant impact on its performance. To reduce the computational burden and the error caused by models not matching the current maneuver, the threshold t_2 should be decreased. However, as the threshold t_2 decreases, the problem of difficulty in activating a model consistent with the current maneuver will gradually arise, also introducing estimation errors. Similarly, decreasing the threshold t_1 will cause more models to be classified as unlikely models and thus be eliminated. In contrast, as the threshold t_1 is increased, a case will arise in which models that should have been eliminated are still retained in the model set to participate in the estimation, increasing the computational burden. Both of these cases lead to increased estimation error. Therefore, the thresholds t_1 and t_2 should be adjusted in accordance with the actual situation to find their balance points.

Notably, we use different kinds of models in the variable-structure multimodel estimation process. The different kinds of models used here are the models described in Section II (VT model,DK model, etc.), which have different structures and represent different maneuvers. In contrast, the traditional VSMM approach uses only one kind of model; for example, if the constant acceleration (CA) model is used the acceleration mode space is quantified by setting different acceleration values to obtain the model set.

It is worth stating that the IMM-MHE algorithm introduces historical time-domain information, which has higher estimation accuracy but increases the computational burden. For this, the VSIMM-MHE algorithm is proposed, which reduces the computational burden and decreases the estimation errors due to mismatched models. Rather than modifying specific steps of the IMM-MHE algorithm, the VSIMM-MHE algorithm incorporates the IMM-MHE algorithm as a part.

Additionally, due to the variable structure of the model set and the introduction of historical time-domain information, analyzing the stability and convergence of the algorithm presents significant challenges. Therefore, this paper verifies the effectiveness and practicality of the proposed algorithm through combined simulation experiments under various

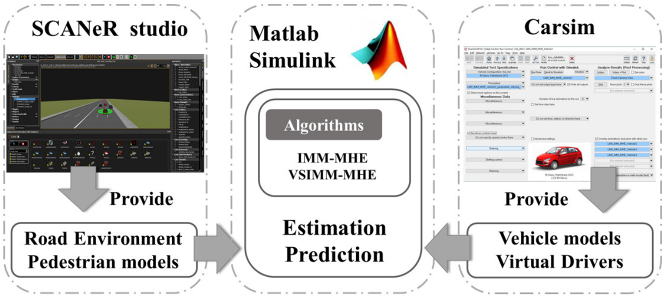


Fig. 6. SCANeR studio, Simulink and Carsim co-simulation.

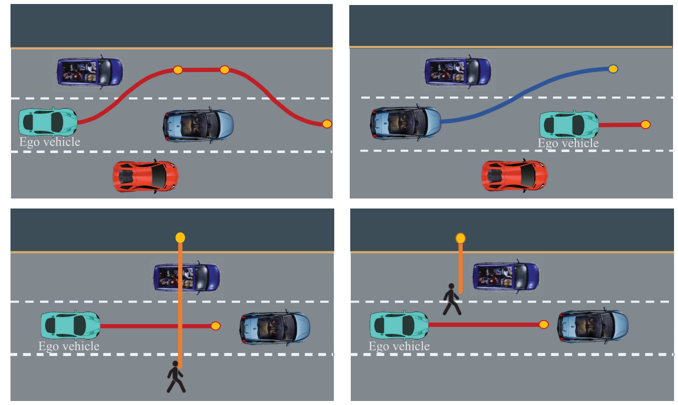


Fig. 7. Simulation scenarios for four operating conditions.

conditions and hardware-in-the-loop experiments. The analysis of the algorithm's stability and convergence will be the focus of our future research work.

V. Simulation Results

In this section, the estimation effect of IMM-MHE, the estimation effect of VSIMM-MHE, and the motion prediction effect based on the model probabilities in VSIMM-MHE are verified by the joint simulation of SCANeR studio, Carsim, and Simulink, as well as by hardware-in-the-loop experiments. Carsim provides the vehicle information, SCANeR studio provides the environment and pedestrian information, and Simulink estimates and predicts the surrounding vehicles based on the estimation algorithm after receiving these data, as shown in Figure 6. The software versions used are SCANeR studio 1.9, MATLAB R2019b, and Carsim 2016.1, and the simulation step size is 0.1 s. Here, a vehicle in the vicinity of the autonomous vehicle is chosen as the reference, referred to as the ego vehicle. In this section, the behavior of other vehicles is described with respect to this ego vehicle.

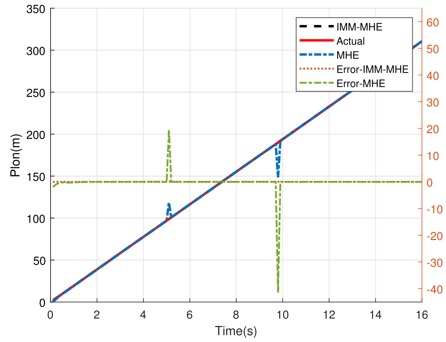
A. IMM-MHE

To verify the validity of the IMM-MHE algorithm, we compare it with the MHE and IMM-KF algorithms under four operating conditions: double lane change, being overtaken, pedestrian approaching and pedestrian staying away.

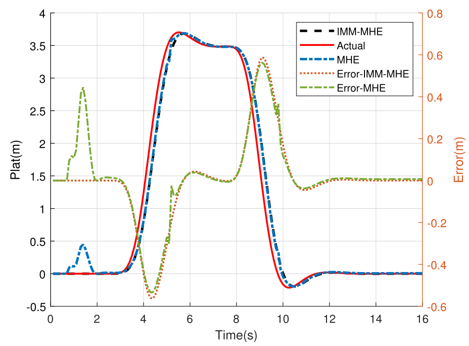
1) Double Lane Change: In reality, it is common for a vehicle to perform a double lane change operation to pass the car in front. As shown in the first scenario in Figure 7, the vehicle in front is driving in the middle lane at a constant velocity of 60 km/h, and the vehicle on the left is driving in the left lane at a constant velocity of 80 km/h. The ego vehicle first changes lanes into the left lane at a velocity of 70 km/h and then returns to the middle lane. Here, model set $\mathcal{M}1$ is chosen.

To demonstrate the inability of a single model to accurately describe the complex motion of a vehicle and the advantages of multimodel estimation, a set of comparative simulation experiments between IMM-MHE and single-model MHE are implemented in this paper, as shown in Figure 8. The model chosen for MHE is the $DK-LC1$ model. The average errors of IMM-MHE on the longitudinal velocity, longitudinal acceleration and lateral position are 0.0103 m/s, 0.0061 m/s^2 and 0.0005 m, respectively, while the corresponding errors of MHE are as high as 0.0926 m/s, 0.1738 m/s^2 and 0.0197 m, respectively. Among them, the errors of MHE before 2 s are more obvious. Meanwhile, since the vehicle fully enters the left lane and returns to the middle lane at 5 s and 9.5 s, respectively, at this time front vehicle of the ego vehicle switches, and the information such as the position and speed of the front vehicle jumps. For the estimators, it is equivalent to the external input increasing the interference. In this situation, the results of MHE exhibit sudden spikes, while the IMM-MHE algorithm is almost unaffected. These phenomena occur because the model corresponding to MHE does not match the real motion of the vehicle well. The IMM-MHE algorithm proposed in this research is a multi-model estimation algorithm that uses model probabilities to perform "soft switching" of models. The more the model matches the current movement of the vehicle, the higher the model probability, and the greater the proportion of the estimated value of the model when finally estimated fusion. Therefore, IMM-MHE has higher estimation accuracy and anti-interference capability. The single-model MHE cannot satisfy the complex driving of vehicles.

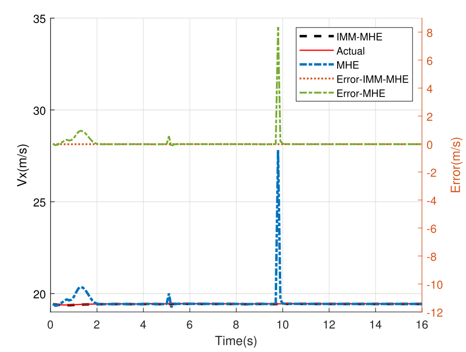
To illustrate the necessity of making full use of historical information, a comparative simulation experiment of the IMM-MHE and IMM-KF algorithms is also presented. As shown in Figure 9, when the parameter matrices Q and R and other conditions are the same, the estimation performance of IMM-MHE is better than that of the IMM-KF. It is worth stating that the Q and R matrices have a significant impact on the estimation results, where the Q matrix represents the level of confidence in the models and the R matrix indicates the level of confidence in the measured data. The Q and R matrices are first adjusted so that the IMM-KF algorithm has a good estimation accuracy, and then the validation of the effectiveness of the IMM-MHE algorithm and the VSIMM-MHE algorithm is performed with the same Q and R matrices. Comparison of the superiority of the proposed algorithms in the case of the models and the same level of trust in the models and measured data. During lane change operations, the IMM-KF has more apparent errors, with average errors of 0.0168 m/s and 0.0012 m on the longitudinal velocity and lateral position, respectively. In comparison, IMM-MHE yields average errors of only 0.0105 m/s and 0.0002 m. $v_{ref,k}$ and $t_{gap,k}$ capture the driver's intentions corresponding to the VT



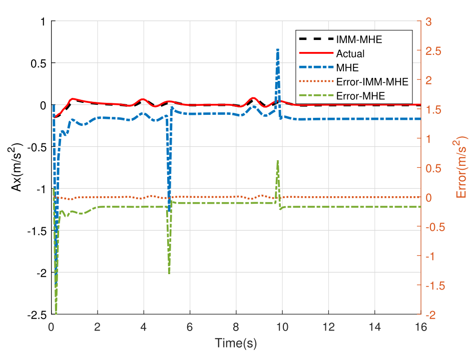
(a) Longitudinal position



(b) Lateral position

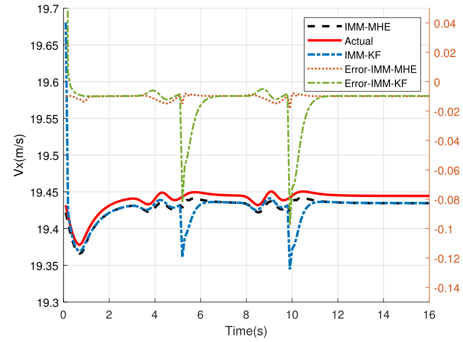


(c) Longitudinal velocity

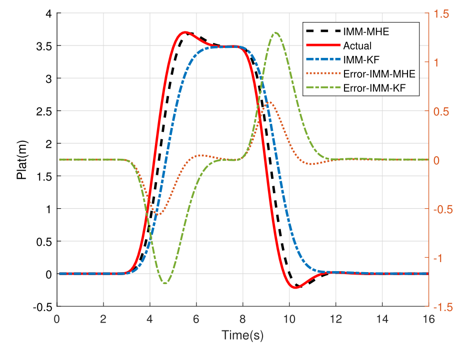


(d) Longitudinal acceleration

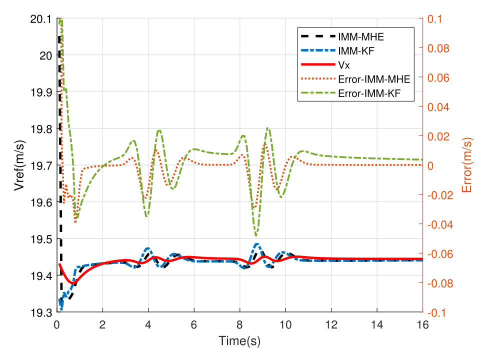
Fig. 8. Comparison results of the estimated performance of IMM-MHE and MHE under double lane change condition.



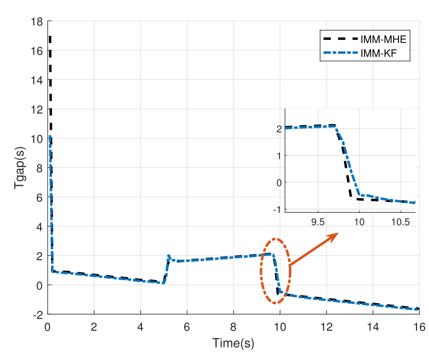
(a) Longitudinal velocity



(b) Lateral position



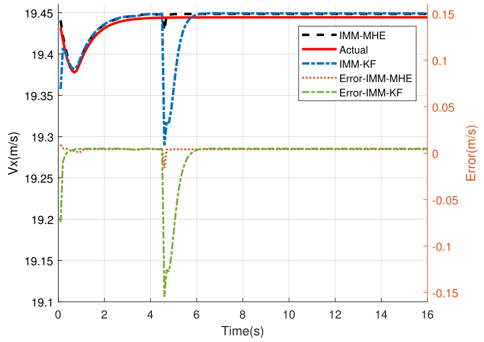
(c) Velocity reference



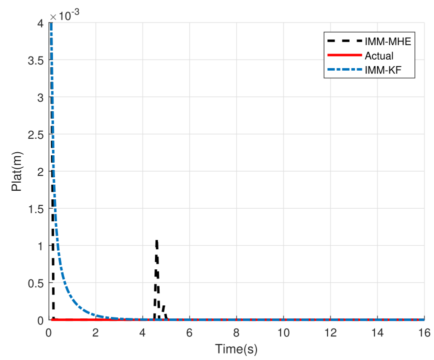
(d) Longitudinal time gap

Fig. 9. Comparison results of the estimated performance of IMM-MHE and IMM-KF under double lane change condition.

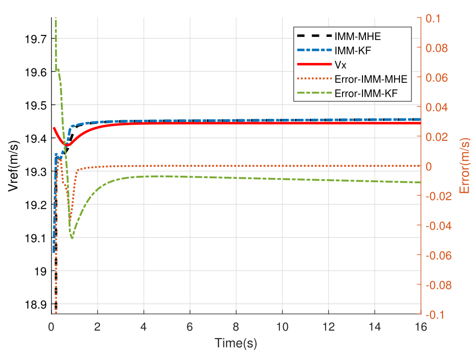
and DK models, respectively, where $v_{ref,k}$ is the longitudinal reference velocity of the vehicle and $t_{gap,k}$ ensures that the distance between the vehicle and the vehicle in front remains constant. The simulation results show that IMM-MHE has a faster response rate for $t_{gap,k}$ estimation. Initially, since the ego vehicle's velocity is greater than that of the vehicle in front, its



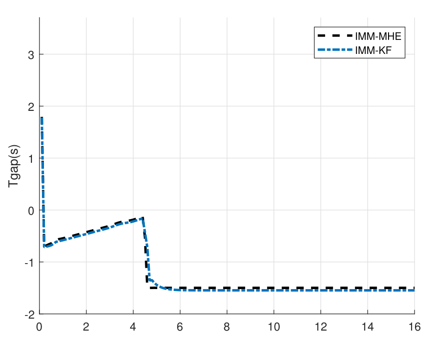
(a) Longitudinal velocity



(b) Lateral position



(c) Velocity reference



(d) Longitudinal time gap

Fig. 10. Comparison results of the estimated performance of IMM-MHE and IMM-KF under being overtaken condition.

velocity continuously decreases but always remains above 0. The ego vehicle completes the lane change into the left lane at 5 s, and the vehicle in front switches to the vehicle that was originally on the left. Consequently, a jump is produced in $t_{gap,k}$ because the distance after the switch is greater than the distance before the switch. At this time, the velocity of the ego vehicle is less than the velocity of the vehicle in front, so $t_{gap,k}$ increases. When the ego vehicle returns to the middle lane at 9.5 s, the vehicle in front switches back to the original vehicle in front. However, at this time, this vehicle is behind the ego vehicle, and its velocity is lower. Therefore, $t_{gap,k}$ jumps to a negative value and then continues to decrease.

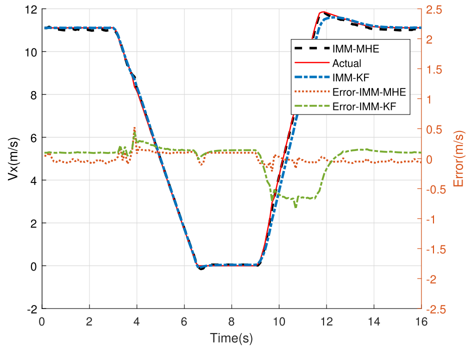
In conclusion, in the double lane change scenario, the IMM-MHE algorithm has higher estimation accuracy and better anti-interference capabilities than the MHE algorithm, and it has higher estimation accuracy and a faster response speed than the IMM-KF algorithm.

2) Being Overtaken: As shown in the second scenario in Figure 7, the ego vehicle is driving in the middle lane at 70 km/h. The vehicle behind changes from the middle lane to the left lane at 80 km/h and then continues driving in the left lane. The vehicle farther behind drives in the middle lane at 70 km/h. As in the previous scenario, model set $\mathcal{M}1$ is selected.

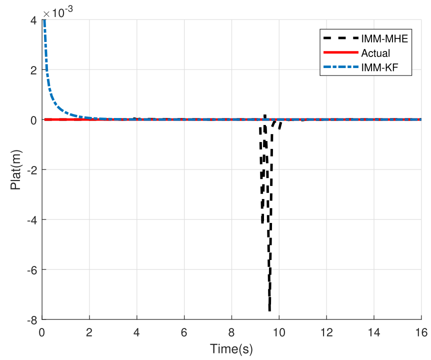
As shown in Figure 10, the estimation performance of IMM-MHE is better than that of the IMM-KF when the parameter matrices Q and R and other conditions are the same. When the ego vehicle is overtaken, the rear vehicle switches, and its position and velocity information jumps, which is equivalent to adding interference to the external inputs, and the longitudinal velocity estimates of both IMM-KF and IMM-MHE are affected, but the IMM-MHE has smaller errors and shorter adjustment times. Specifically, IMM-MHE achieves a much shorter response time for $t_{gap,k}$ estimation when the rear vehicle switches.

3) Pedestrian Approaching: During daily driving, one may often encounter obstacles ahead or pedestrians crossing the road. When it is impossible to change lanes due to the restrictions of the surrounding environment or when merely slowing down cannot guarantee the safe passage of pedestrians, drivers should stop and give way. In the scenario considered here, the ego vehicle is driving in the middle lane at 40 km/h when the driver notices that a pedestrian is crossing the road and has not reached the center lane, so he stops to give way. After the pedestrian has passed, the driver initially accelerates back to 40 km/h and then drives at a constant velocity. The vehicle in front continues driving in the middle road at 40 km/h, as shown in the third scenario in Figure 7. Unlike in the first two operating conditions, model set $\mathcal{M}2$ is selected here.

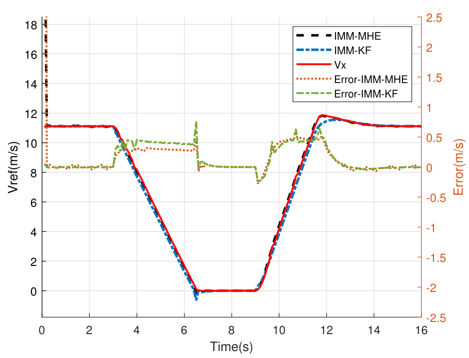
Figure 11 shows that when the parameter matrices Q and R and other conditions are the same, the estimation performance of IMM-MHE is better than that of IMM-KF. From the start of vehicle deceleration to the recovery to the original velocity, the longitudinal velocity estimates of IMM-MHE have an average error of only 0.0005 m/s, while the average error of the IMM-KF is 0.1643 m/s. Although IMM-MHE briefly generates some peaks during switching, these peaks are small in magnitude, so it is considered equivalent to the magnitude of the lateral position estimation error of the IMM-KF. For the estimation of the extended variables $v_{ref,k}$ and $d_{pre,k}$,



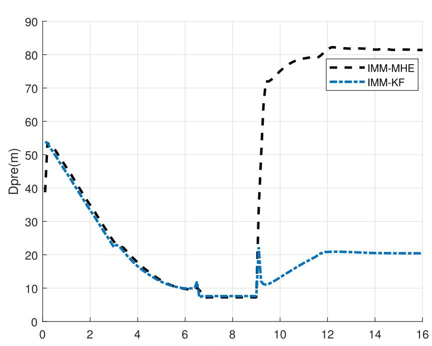
(a) Longitudinal velocity



(b) Lateral position

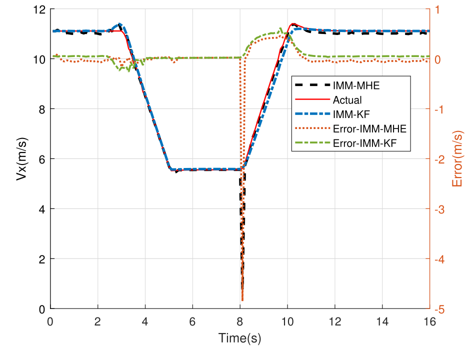


(c) Velocity reference

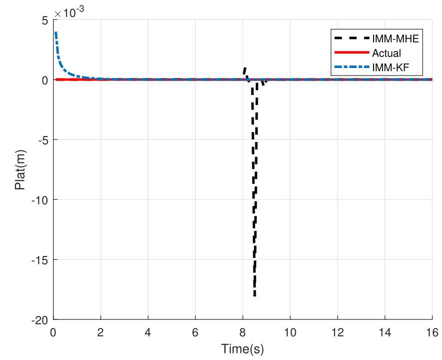


(d) Separation distance

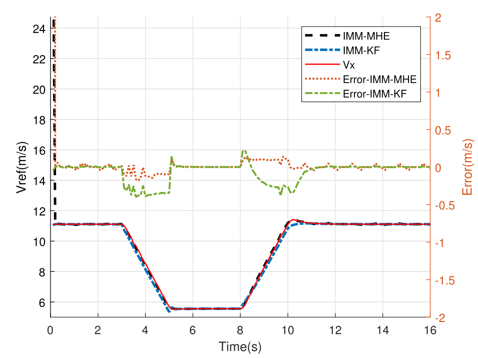
Fig. 11. Comparison results of the estimated performance of IMM-MHE and IMM-KF under pedestrian approaching condition.



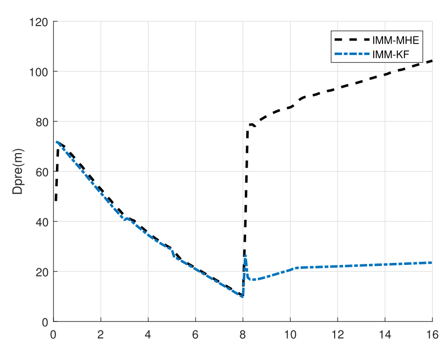
(a) Longitudinal velocity



(b) Lateral position



(c) Velocity reference



(d) Separation distance

Fig. 12. Comparison results of the estimated performance of IMM-MHE and IMM-KF under pedestrian staying away condition.

IMM-MHE has a clear advantage, especially for $d_{pre,k}$. The IMM-KF's estimate of $d_{pre,k}$ is incorrect after the switch.

4) Pedestrian Staying Away: When driving in the middle lane at 40 km/h, the driver notices a pedestrian crossing the

TABLE II
THE AVERAGE ERROR OF v_{ref} (m/s)

Operating condition	**IMM-MHE**	**IMM-KF**
Double lane change	0.0023	0.0033
Overtaken	0.0095	0.0105
Pedestrian approaching	0.0037	0.1537
Pedestrian staying away	0.0008	0.0717

road who is about to pass the center lane, so he begins to slow down and pass at a low velocity. After the pedestrian passes, the driver resumes driving at his original velocity, as shown in the fourth scenario in Figure 7. Meanwhile, the vehicle in front drives in the middle lane at a speed of 50 km/h. As in the previous scenario, model set $\mathcal{M}2$ is selected.

Figure 12 again shows that IMM-MHE yields better estimates than the IMM-KF when the parameter matrices Q and R and other conditions are the same. During the time from vehicle deceleration to velocity recovery, although the jump in information such as position and speed caused by the switch from the pedestrian in front to the vehicle in front causes a peak in the longitudinal velocity estimate of the IMM-MHE, it completes the correction in a very short time and the average error remains relatively small at 0.0326 m/s. In contrast, the average error of the IMM-KF for longitudinal velocity estimation is 0.0614 m/s. Similar to the pedestrian approaching scenario, the $d_{pre,k}$ estimate generated by the IMM-KF shows a significant discrepancy from the true value after the switch is completed.

Table II gives the average errors of the $v_{ref,k}$ estimates obtained by the IMM-MHE and IMM-KF algorithms under the four operating conditions above. The data show that the average errors of IMM-MHE for $v_{ref,k}$ estimation are all smaller than those of the IMM-KF, indicating that IMM-MHE has a better estimation effect.

In conclusion, the IMM-MHE algorithm proposed in this paper has a better estimation effect than the IMM-KF and MHE algorithms in both estimation accuracy and convergence speed.

B. VSIMM-MHE

To illustrate the effectiveness of the VSIMM-MHE algorithm, we present additional simulation experiments performed under the above four operating conditions, which are divided into two main parts, namely, model set change and model set fixation.

In addition, to provide better information support for trajectory planning for autonomous vehicles, the maximum model probability obtained in the VSIMM-MHE algorithm is used to predict the motion trajectories of the surrounding vehicles in this paper. The maximum model probability can be used to identify the motion model that best matches a vehicle's current driving behavior, and because the vehicle motion is coherent, the identified model can be used to predict the vehicle's motion trajectory in the short term. The process of calculating the maximum model probability in VSIMM-MHE is divided into two stages. The first stage is the identification of the model set, first screening out models that clearly do not match the current maneuver to reduce the error. The second stage is multimodel estimation after the model set is identified. Equation (23) is used to obtain the model probabilities of all models in the model set at the current timestep, the largest of which corresponds to the model that best matches the current maneuver. Subsequently, the trajectories of the surrounding vehicles are calculated using the respective models. It is worth stating that the selection of the prediction time length is not related to the selection of the time-domain length of the MHE.

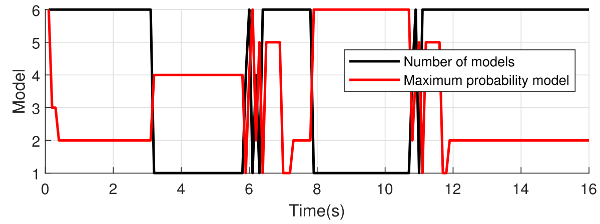


Fig. 13. Number of models and maximum probability model for VSIMM-MHE under double lane change condition.

1) Model Set Change: The VSIMM-MHE algorithm maintains six models while the vehicle is driving in a straight line to consider the possibility of three maneuvers that can occur directly in the next timestep: staying in the current lane, switching lanes to the left, or switching lanes to the right. This prevents the estimation error from increasing due to a long switching process of model sets, which may lead to subsequent safety issues, etc. While the vehicle is performing a lane change, however, motion models that do not match the current maneuver of the vehicle are discarded and the model set changes. Here, the double lane change condition described above is chosen.

Figure 13 shows the number of models in the model set and the maximum probability model, where the vertical coordinates of the maximum-probability model curve correspond to models 1–6. Models 1 and 4 correspond to switching lanes to the left, models 2 and 5 correspond to staying in the current lane, and models 3 and 6 correspond to switch lanes to the right. As shown in Figure 13, the number of models in the model set switches from 6 to 1 when the vehicle is changing lanes and then changes back to 6 after the lane change is completed, achieving a reduction in the computational burden of the estimator. Meanwhile, the maximum-probability model curve illustrates that VSIMM-MHE can well identify the best matching model for the vehicle during driving. From this curve, it can be seen that the vehicle is going straight, performing a left lane change, going straight, performing a right lane change and going straight in sequence, which is consistent with the double lane change working condition. Notably, the oscillations generated after a lane change are the result of the combined effects of the linear quadratic regulator and the model set switching process. As seen from the above, the VSIMM-MHE algorithm can reduce the computational pressure while matching a vehicle's maneuvers.

In addition, Table III compares the estimation accuracies of VSIMM-MHE and IMM-MHE when the model set changes. Here, the mean absolute error (MAE), root mean square Error (RMSE), and mean absolute percentage error (MAPE) are

TABLE III
COMPARISON RESULTS OF THE ESTIMATION ACCURACY OF VSIMM-MHE AND IMM-MHE WHEN THE MODEL SET CHANGES

		VSIMM-MHE	IMM-MHE
V_x	MAE	0.0039	0.0117
	RMSE	0.0052	0.0118
	MAPE	0.0199	0.0600
P_{lat}	MAE	0.3485	0.3557
	RMSE	0.3959	0.4002
	MAPE	44.2173	46.9022

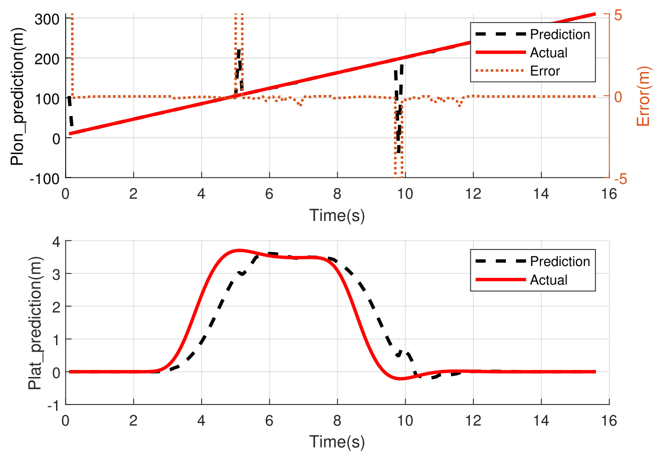


Fig. 14. Longitudinal position prediction and lateral position prediction for VSIMM-MHE under double lane change condition.

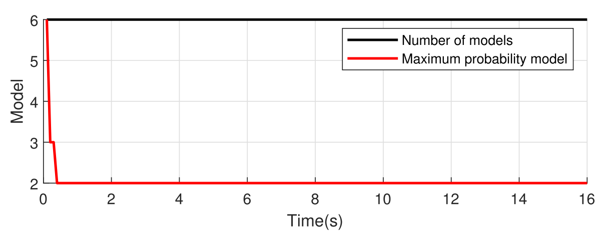


Fig. 15. Number of models and maximum probability model for VSIMM-MHE under being overtaken condition.

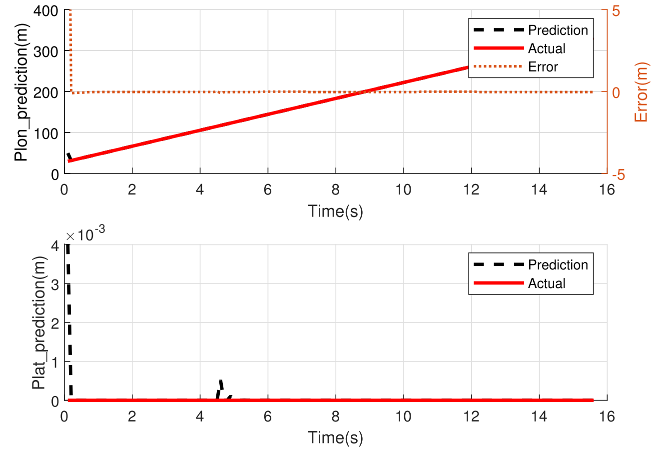


Fig. 16. Longitudinal position prediction and lateral position prediction for VSIMM-MHE under being overtaken condition.

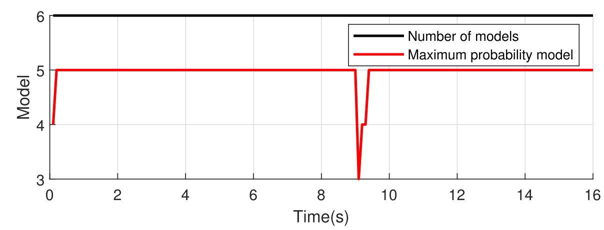


Fig. 17. Number of models and maximum probability model for VSIMM-MHE under pedestrian approaching condition.

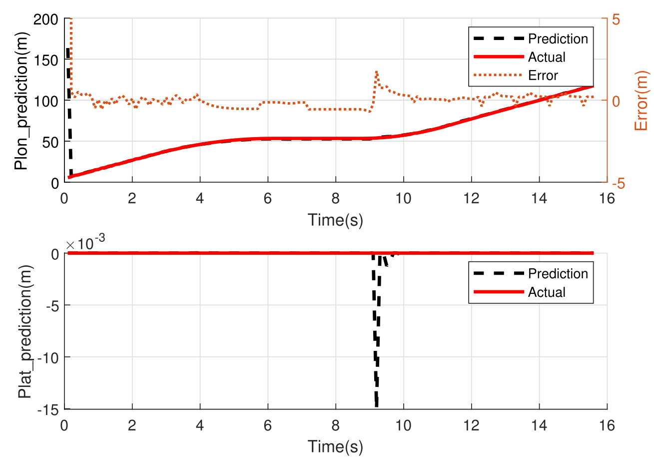


Fig. 18. Longitudinal position prediction and lateral position prediction for VSIMM-MHE under pedestrian approaching condition.

chosen as the evaluation indicators. As shown in the table, the estimation errors of VSIMM-MHE are smaller. Compared with the IMM-MHE algorithm, VSIMM-MHE can eliminate the estimation error due to models that do not match the current maneuver, thereby improving the final estimation accuracy.

Figure 14 illustrates that the model with the maximum model probability in VSIMM-MHE yields excellent results for motion trajectory prediction, and although the switching of vehicles in front leads to spikes in the predicted values, convergence can be achieved in a short time.

2) Model Set Fixation: Under the three operating conditions of being overtaken, pedestrian approaching and pedestrian staying away, the ego vehicle is driven in a straight line, and all six models are maintained in the model set. The number of models and the maximum-probability model corresponding to each of these three operating conditions are shown in Figures 15, 17 and 19, respectively. VSIMM-MHE can effectively identify the model that best matches the current maneuver during the driving process. In Figure 17, the maximum-probability model jumps to model 3 at 9 s, which is due to the switch from the pedestrian in front to the vehicle in front at that timestep; however, recovery can be achieved in a very short time. Figures 16, 18 and 20 show the vehicle trajectories presented using the maximum probability model of VSIMM-MHE for the three operating conditions, respectively. The results show that the proposed method has a good prediction effect, and although a large prediction error occurs when the front vehicle or pedestrian switches, the effect of the switch disappears in a short time.

In summary, the VSIMM-MHE algorithm not only reduces the computational burden, but also eliminates the estimation error caused by model mismatch with respect to the current maneuver, and the maximum model probability can be

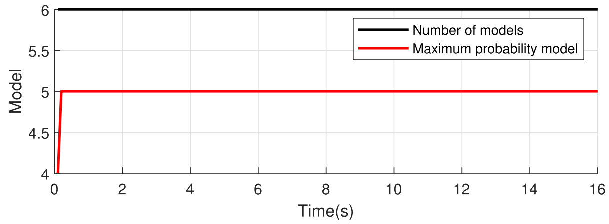


Fig. 19. Number of models and maximum probability model for VSIMM-MHE under pedestrian staying away condition.

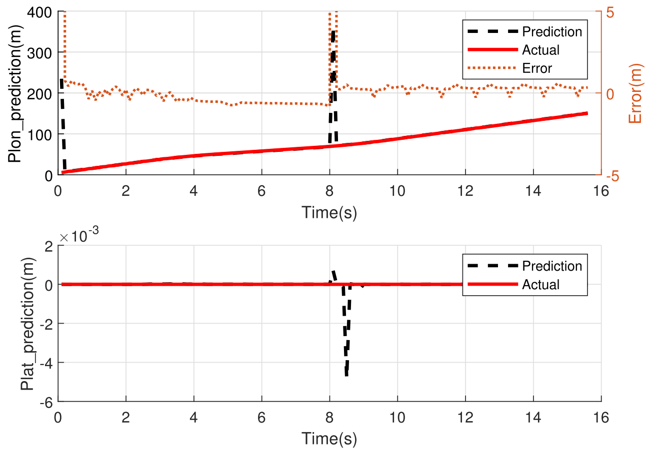


Fig. 20. Longitudinal position prediction and lateral position prediction for VSIMM-MHE under pedestrian staying away condition.

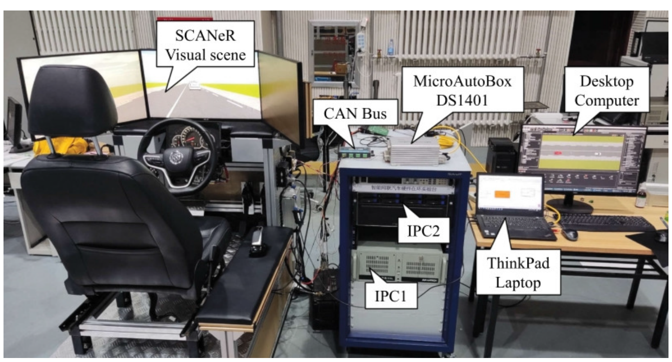
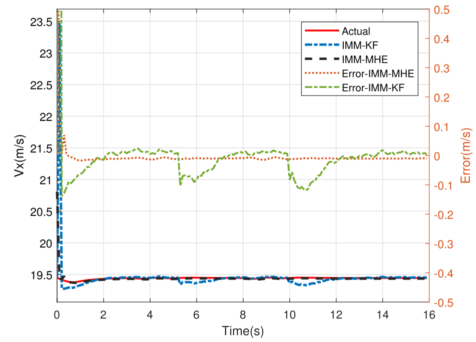


Fig. 21. Platform of the HiL experiment.

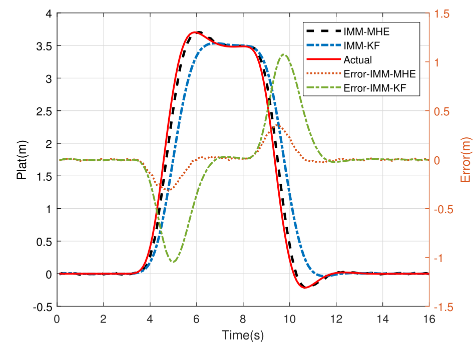
effectively utilized to predict the future motion of the vehicle over a short-term period.

C. Hardware-in-the-Loop Experimental Results

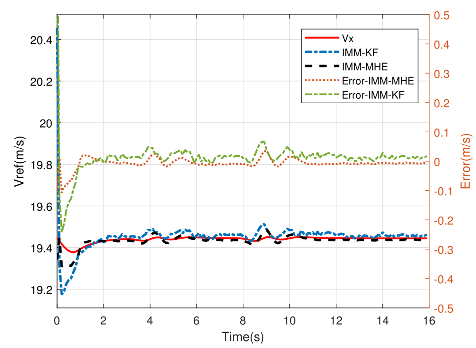
Hardware-in-the-loop (HiL) experiments are conducted to further validate the effectiveness of the proposed algorithms. The HiL platform consists of two industrial personal computers (IPCs), one laptop, one desktop computer, a Controller Area Network (CAN) bus, and a dSPACE MicroAutoBox DS1401, as shown in Figure 21. The IPC1 and desktop computer are used for vehicle dynamics simulation with CarSim as the vehicle dynamics software, while IPC2 provides a simulated driving environment using SCANeR studio. The vehicles' double lane change and straight-line movements are provided by CarSim and not through driver inputs such as steering wheel and throttle. The MicroAutoBox is used as a rapid prototype estimator, and the laptop is used to monitor the entire system and store data. Additionally, the CAN bus is used for data transmission between the aforementioned devices.



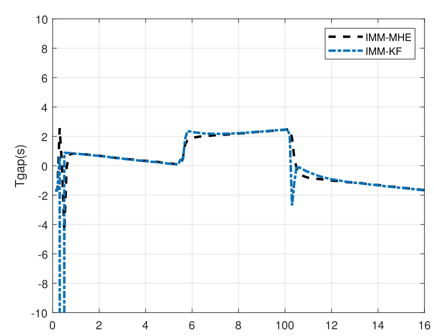
(a) Longitudinal velocity



(b) Lateral position



(c) Velocity reference



(d) Longitudinal time gap

Fig. 22. Comparative results of the estimation performance of IMM-MHE and IMM-KF under hardware-in-the-loop for double lane change condition.

TABLE IV

COMPARISON RESULTS OF THE ESTIMATION ACCURACY OF THE VSIMM-MHE AND IMM-MHE ALGORITHMS UNDER THE HARDWARE-IN-THE-LOOP WHEN THE MODEL SET IS CHANGED

		VSIMM-MHE	**IMM-MHE**
V_x	MAE	0.0031	0.0095
	RMSE	0.0076	0.0099
	MAPE	0.0161	0.0491
P_{lat}	MAE	0.1814	0.1970
	RMSE	0.1982	0.2288
	MAPE	23.4172	26.3643

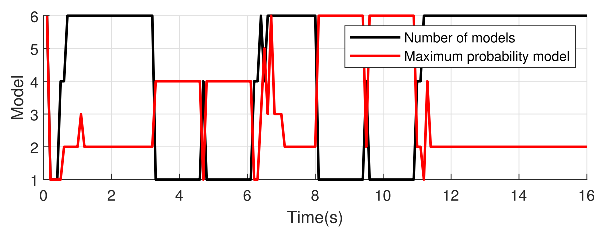


Fig. 23. Number of models and maximum probability model for VSIMM-MHE under double lane change condition.

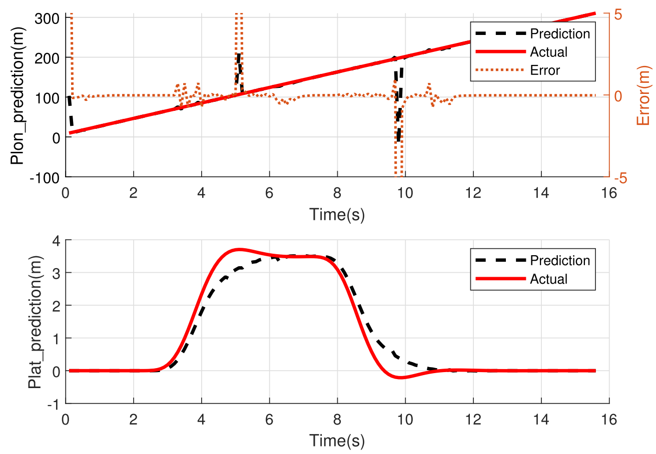


Fig. 24. Longitudinal position prediction and lateral position prediction for VSIMM-MHE under hardware-in-the-loop for double lane change conditions.

To show the variable structure of the model set of the VSIMM-MHE algorithm, the double lane change condition is selected, and the scenario setup is the same as that of Section V-A.1. The simulation step size is also 0.1 s.

Figure 22 shows a comparison of the estimation results of the IMM-MHE algorithm and the IMM-KF algorithm for the double lane change condition in the hardware-in-the-loop experiment. The results similarly show that IMM-MHE makes full use of historical time-domain information and has higher estimation accuracy. Table IV further illustrates that the VSIMM-MHE algorithm achieves a reduction in estimation error due to the elimination of poorly matched models. From Figure 23, the VSIMM-MHE algorithm can effectively implement the variable structure of the model set and reduce the computational burden. Meanwhile, the model that best matches the vehicle's current maneuver can be effectively identified by the maximum model probability in the VSIMM-MHE algorithm. And Figure 24 shows that the vehicle's motion trajectory over a short-term period in the future can be effectively predicted using this model.

VI. CONCLUSION

In this paper, an estimation scheme is proposed for estimating the state information of the vehicles in the vicinity of an autonomous vehicle. First, the IMM-MHE algorithm is proposed, which can match multiple vehicle maneuvers while fully utilizing historical information. Second, the VSIMM-MHE framework is designed. In this framework, the models in the model set are classified into principle models, significant models and unlikely models using a new model classification method. In each iteration, the models adjacent to the principal models are activated, the principle and significant models are retained, and some less likely models are eliminated. In addition, time-domain adaptation and an interaction method are designed for variable structure handing. Third, the VSIMM-MHE algorithm is proposed for model sets consisting of different kinds of intention models, which reduces the dependence on the accuracy of the model probabilities through residual information and utilizes the model probabilities in this algorithm to perform motion prediction for surrounding vehicles. A series of simulation experiments and a hardware-in-the-loop experiment were conducted to verify the validity of the proposed estimation algorithm and their predictions. The experimental results show that the IMM-MHE algorithm offers higher estimation accuracy and faster convergence speed than the MHE and IMM-KF algorithms. The VSIMM-MHE algorithm can adjust the model set in accordance with the actual driving behavior of a vehicle, with a reduced computational burden and higher estimation accuracy. In addition, the VSIMM-MHE algorithm can effectively identify the model that most closely matches the current actual maneuver, and then accurately predict the future motion of the vehicle.

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